SUPPLEMENTAL MATERIAL

Table S1. Search strategy used in the current systematic review and meta-analysis.

Meniscus	1. menisc*
	2. meniscal repair
	3. meniscus repair
	4. meniscal surgery
	5. Meniscus [MeSH]
	6. OR 1-6
Second-look arthroscopy	7. second-look arthroscopy
	8. second-look arthroscopic
	9. repeat arthroscopy
	10. repeat arthroscopic
	11. Second-Look Surgery [MeSH]
	12. OR 7-11
Combined search	6 AND 12

MeSH, Medical Subject Heading.

Table S2. Definition of partial healing under second-look arthroscopy in each included study.

Study	Definition of partial healing
Jing[26] (2020)	An apparent increase in meniscus lifting and mobility on probing with good meniscal continuity.
DePhillipo[17] (2019)	—
Kanto[30] (2019)	A partial defect of 50% of the original repair length or height that was stable to probing.
Kaminski[28] (2019)	There was at least 50% healing of the tear width and stable repair.
Lee[36] (2019)	Apparent increases in meniscus lifting and mobility on probing with good meniscal continuity.
Seo[55] (2019)	The residual cleft residual cleft was less than 50%.
Song[59] (2019)	There was no obvious penetration, and the probe could only be partially inserted.
Tsujii[69] (2019)	If a partial-thickness defect was visible, it was classified as partial healing.
Chen[13] (2018)	Cases where the repair site exhibited satisfactory stability but a partial defect ≥ 50% of the original repair length or height was visible.
Lee[37] (2018)	—
Tsujii[68] (2018)	If there was a small defect with synovial coverage over more than half of the torn area, it was classified as partial healing.
Matsushita[39] (2017)	A partial defect of 50% of the original repair length or height that was stable to probing.
Park[47] (2017)	—
Ahn[2] (2015)	In the partial healing group, the unhealed cleft of tear had to be less than 30% of the whole length of the initial horizontal cleavage tear.
Kang[29] (2015)	Residual cleft at the repair site of the meniscus was less than 50%.
Ra[48] (2015)	—
Cho[14] (2014a)	—
Cho[15] (2014b)	Apparent increase in meniscus lifting and mobility on probing with good meniscal continuity
Hagino[21] (2014)	—
Song[60] (2014)	A partially healed meniscus was characterized by a residual cleft at the tear site of less than 50% of the thickness of the meniscus.
Ra[49] (2013)	Healing of 90–50 % was incomplete.
Ruiz-Iban[51] (2012)	—
Kim[31] (2011)	A case with slight motion or subluxation was referred to as incomplete healing.
Miao[41] (2011)	—
Seo[54] (2011)	Apparent increase in meniscus lifting and mobility on probing with good meniscal continuity
Ahn[3] (2010)	A residual cleft at the tear site of less than 50% (1-3 mm of the inserted probe tip) full thickness of the meniscus.
Suganuma[63] (2010)	Stable bonding at the repair site but with evidence of some partial-thickness defect of the surface.
Tachibana[65] (2010)	A tear that was healed over its full length with a residual cleft that was less than 50% of its vertical height was classified as incompletely healed.
Lee[35] (2009)	—
Feng[20] (2008)	A partial defect of less than 50% of the original repair length or height that was stable to probing.
Sarimo[53] (2005)	—
Soejima[58] (2005)	Partial thickness defect.
Asik[8] (2002)	—
Kurosaka[34] (2002)	—
Albrecht-Olsen[5] (1999)	—
van Trommel[70] (1998)	—
Asahina[7] (1996)	A partial defect of less than 50% of the original repair length or height that was stable to probing.
Horibe[24] (1995)	Restoration of the stability of the repaired site although a partial-thickness defect was visible.
Kimura[32] (1995)	—
Tenuta[67] (1994)	—
Morgan[43] (1991)	A partial defect of less than 50% of the original repair length or height that was stable to probing.

Table S3. Modified Newcastle-Ottawa Scoring for Observational Studies

Representativeness of the sample:
1 point Population contained a mixture of specialties at multiple sites.
0 points Population contained a single specialty at a single site.
Sample size:
1 point Sample size was greater than 200 participants.
0 points Sample size was less than 200 participants.
Comparability:
1 point The comparability between characteristics of each group was established and satisfactory.
0 points The comparability between characteristics of each group was unsatisfactory, or there was no description of the characteristics of each group.
Ascertainment of meniscal healing status:
1 point Validated measurement tool, or using a validated cutoff score.
0 points Non-validated measurement tool, or validated measurement tool with non-valid cutoff score.
Quality of descriptive statistics reporting:
1 point Reported descriptive statistics to describe the population (eg, age, sex) with proper measures of dispersion (eg, standard deviation, standard error, range).
0 points Descriptive statistics were not reported, were incomplete, or did not include proper measures of dispersion.

Studies were judged to be of low risk of bias (≥3 points) or high risk of bias (<3 points).

Table S4. Quality assessment of the included studies

Observational studies	Representativeness	Sample size	Comparability	Ascertainment of meniscal healing status	Quality of statistics reporting	Total modified Newcastle-Ottawa Score
Lee[35] (2009)	0	0	0	1	1	2
Kang[29] (2015)	0	0	0	1	1	2
Cho[14] (2014a)	0	0	0	1	1	2
Cho[15] (2014b)	0	0	0	1	1	2
van Trommel[70] (1998)	0	0	0	1	1	2
Tsujii[69] (2019)	0	0	1	1	1	3
Tsujii[68] (2018)	0	0	0	1	1	2
Tenuta[67] (1994)	0	0	0	1	0	1
Tachibana[65] (2010)	0	0	0	1	1	2
Song[60] (2014)	0	0	0	1	1	2
Song[59] (2019)	0	0	0	1	1	2
Soejima[58] (2005)	0	0	0	1	0	1
Seo[54] (2011)	0	0	0	1	1	2
Sarimo[53] (2005)	1	0	0	1	1	3
Ruiz-Iban[51] (2012)	0	0	0	1	1	2
Ra[49] (2013)	0	0	0	1	1	2
Ra[48] (2015)	0	0	0	1	1	2
Morgan[43] (1991)	0	0	0	1	1	2
Matsushita[39] (2017)	0	0	0	1	1	2
Lee[37] (2018)	0	0	0	1	1	2
Kurosaka[34] (2002)	0	0	0	1	1	2
Kimura[32] (1995)	0	0	0	1	1	2
Kim[31] (2011)	0	0	1	1	1	3
Kanto[30] (2019)	0	0	0	1	1	2
Horibe[24] (1995)	0	0	0	1	0	1
Hagino[21] (2014)	0	0	0	1	1	2
Feng[20] (2008)	0	0	0	1	1	2
Chen[13] (2018)	0	0	0	1	1	2
Asik[8] (2002)	1	0	0	1	1	3
Asahina[7] (1996)	0	0	0	1	1	2
Ahn[3] (2010)	0	0	0	1	1	2
Ahn[2] (2015)	0	0	0	1	1	2
Park[47] (2017)	0	0	0	1	1	2
DePhillipo[17] (2019)	0	0	0	1	1	2
Seo[55] (2019)	0	0	1	1	1	3
Jing[26] (2020)	0	0	0	1	1	2
Miao[41] (2011)	0	0	1	1	1	3
Lee[36] (2019)	0	0	0	1	1	2
Suganuma[63] (2010)	0	0	1	1	1	3

Randomized controlled trials[b]	Sequence generation	Allocation concealment	Blinding of patients and study personnel	Blinding of outcome assessment	No incomplete outcome data	No selective outcome reporting	No other potential threats to validity
Kaminski[28] (2019)	+	+	+	+	+	+	+
Albrecht-Olsen[5] (1999)	+	–	–	–	+	+	+

[b]Assessed with the Cochrane Risk of Bias tool.

Table S5. Sensitivity analysis for the overall healing rates.

Study	Healing rate (%)	Lower 95% CI	Upper 95% CI	tau^2	I^2
Kaminski[28] (2019)	74	67	80	1.01	89%
Lee[35] (2009)	73	66	79	0.91	88%
Kang[29] (2015)	73	66	79	0.92	88%
Cho[14] (2014a)	73	66	79	0.86	88%
van Trommel[70] (1998)	73	66	79	0.94	88%
Tsujii[69] (2019)	74	67	80	1.00	89%
Tsujii[68] (2018)	75	68	80	0.83	87%
Tenuta[67] (1994)	74	67	80	1.01	88%
Tachibana[65] (2010)	74	66	80	1.02	89%
Song[60] (2014)	74	67	80	0.99	89%
Song[59] (2019)	73	66	79	0.93	88%
Soejima[58] (2005)	74	67	80	0.99	89%
Seo[54] (2011)	75	68	80	0.72	85%
Sarimo[53] (2005)	73	66	80	0.97	88%
Ruiz-Iban[51] (2012)	73	66	79	0.94	88%
Ra[49] (2013)	74	66	80	0.98	89%
Ra[48] (2015)	73	66	79	0.93	88%
Morgan[43] (1991)	74	67	80	1.01	88%
Miao[41] (2011)	73	66	80	0.97	88%
Matsushita[39] (2017)	74	67	80	0.99	88%
Lee[37] (2018)	74	67	80	1.01	89%
Kurosaka[34] (2002)	74	66	80	1.01	88%
Kimura[32] (1995)	73	66	80	0.99	89%
Kim[31] (2011)	74	67	80	1.00	89%
Kanto[30] (2019)	74	67	80	0.99	88%
Horibe[24] (1995)	74	66	80	1.02	88%
Hagino[21] (2014)	74	66	80	1.01	89%
Feng[20] (2008)	74	66	80	0.99	88%
Cho[15] (2014b)	74	68	80	0.90	88%
Chen[13] (2018)	73	66	80	0.97	88%
Asik[8] (2002)	73	66	80	0.99	89%
Asahina[7] (1996)	74	66	80	1.02	88%
Albrecht-Olsen[5] (1999)	74	67	80	1.02	88%
Ahn[3] (2010)	73	66	80	0.99	88%
Ahn[2] (2015)	74	66	80	1.00	89%
Park[47] (2017)	73	66	79	0.90	88%
Lee[36] (2019)	74	67	80	0.93	88%
DePhillipo[17] (2019)	73	66	79	0.95	88%
Seo[55] (2019)	74	67	80	1.01	88%
Jing[26] (2020)	74	67	80	0.93	88%
Suganuma[63] (2010)	74	68	80	0.91	88%

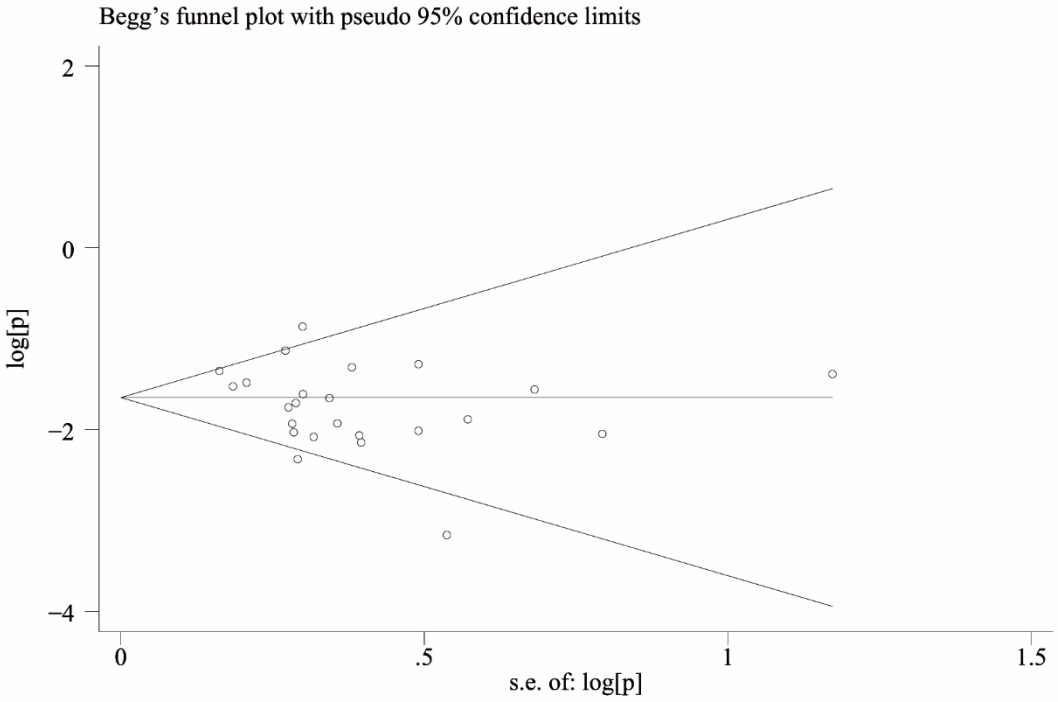


Figure S1. Assessment of small study effects by funnel plot.